Backup and Disaster Recovery Plan

Project: University Library Analytics Data Warehouse

1. Backup Strategy

To prevent data loss, we have designed a three-tiered backup strategy for the MySQL database:

- **Full Backup:** A complete `mysqldump` of the entire `library_dw` database is executed every Sunday at 2:00 AM. This file is compressed and stored on a secure cloud server (e.g., AWS S3 or Google Drive).
- **Incremental Backup:** Every day at 2:00 AM, the MySQL binary logs are backed up to capture only the changes made since the last full backup. This saves storage space and time.
- **Transaction Logs:** The database is configured to write-ahead log (WAL) all transactions, which can be replayed in the event of a sudden crash.

2. Disaster Recovery Plan

In the event of a system failure or data corruption, the following recovery procedure will be executed:

1. **Isolate:** Immediately disconnect the affected server from the network to prevent further corruption.
2. **Restore Full Backup:** Load the most recent Sunday full backup into a clean MySQL instance.
3. **Replay Logs:** Apply the daily incremental backups and binary logs up to the exact minute before the failure occurred.
4. **Verify:** Run data validation checks (e.g., `COUNT(*)` on the fact table) to ensure no data was lost.
5. **Resume:** Redirect the Streamlit dashboard to the restored database and resume operations.

3. Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO)

- **RTO:** The system is designed to be fully restored and operational within **2 hours** of a failure.
- **RPO:** Because of our daily incremental backups and binary logs, the maximum amount of data we can afford to lose is **24 hours** (from the previous night's backup).